

KYUSEONG CHOI



Cornell Tech



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G in

EMPLOYMENT

Member of Technical Staff Intern (Research), Traversal

May 2026–

* Built post-training pipelines to enhance in-house agentic capabilities of open-source models—improved efficiency $\sim 2\times$ (halved tool calls) and improved accuracy by an additional $\sim 10\%$ from baseline.
* Applied off-policy optimization (SFT, DPO) on large ($\sim 1T$) open-source MoE models and on-policy self distillation.

Applied Scientist Intern, Amazon, Pricing & Promotions Science

May–Aug 2025

* Designed kernel-based uniform inference procedures for conditional average treatment effects under endogeneity, informing pricing and promotion policy decisions.

EDUCATION

Ph.D., Statistics and Data Science, Cornell Tech, New York

2021–

Advisors: *Prof. Raaz Dwivedi & Prof. Kengo Kato*

M.S. Biostatistics, University of Michigan, Ann Arbor

2021

Advisors: *Prof. Jeremy Taylor & Prof. Peisong Han*

B.B.A. Business Administration, Korea University, Seoul

2019

Military service (Republic of Korea Air Force)

2014–2016

SELECTED WORKS

Kyuseong Choi, Dwaipayan Saha, Woojeong Kim, Anish Agarwal, Raaz Dwivedi, “GOPO: Policy Optimization using Ranked Rewards”, *COLM, Scientific understanding of Foundational Models (SciFM)*. 2026

Rank-based alternative to GRPO with robustness to reward model noise.

Albert Gong, **Kyuseong Choi**, Abhineet Agarwal, Anish Agarwal, Raaz Dwivedi, “ORCA-Bench: Are Code Agents Ready for Oncall?”. *COLM, Context Beyond the Window (CBW)*. 2026

Benchmark evaluating code agents on production incident response and site-reliability tasks.

Kyuseong Choi, Jacob Feitelberg, Anish Agarwal, Raaz Dwivedi, “Learning counterfactual distributions via kernel methods”, *arXiv, reject & resubmit at Annals of Statistics*. 2024

Nearest-neighbor kernel estimator recovering full counterfactual multivariate distributions

Jacob Feitelberg, Dwaipayan Saha, **Kyuseong Choi**, Anish Agarwal, Raaz Dwivedi, “TabImpute: Accurate and Fast Zero-Shot Missing-Data Imputation with a Pre-Trained Transformer”, *AITD EurIPS*. 2025

WORK IN PROGRESS

Kyuseong Choi, Kengo Kato, “Gaussian approximation in Reproducing Kernel Hilbert spaces, applications to Kernel Ridge Regression”. 2026

PREPRINTS

Kyuseong Choi, Gabriela Ciolek, “Wild regenerative block bootstrap for Harris recurrent Markov chains”, *arXiv, submitted*. 2025

Caleb Chin, Aashish Khubchandani, Harshvardhan Maskara, **Kyuseong Choi**, Jacob Feitelberg, Albert Gong, Manit Paul, Tathagata Sadhukhan, Anish Agarwal, Raaz Dwivedi, “N²: A Unified Python Package and Test Bench for Nearest Neighbor-Based Matrix Completion”, *ICML, CODEML*. 2025

Jacob Feitelberg, Dwaipayan Saha, **Kyuseong Choi**, Zaid Ahmad, Anish Agarwal, Raaz Dwivedi, “TabImpute: Accurate and Fast Zero-Shot Missing-Data Imputation with a Pre-Trained Transformer”, *arXiv, submitted*. 2025

Jacob Feitelberg, **Kyuseong Choi**, Anish Agarwal, Raaz Dwivedi, “Distributional matrix completion via nearest neighbors in the Wasserstein space”, *arXiv*. 2024

PUBLICATIONS	Albert Gong, Kyuseong Choi , Raaz Dwivedi, “Supervised kernel thinning”, <i>NeurIPS</i> . 2024
	Kyuseong Choi , Jeremy Taylor, Peisong Han, “Robust data integration from multiple external sources for generalized linear models with binary outcomes”, <i>Biometrics</i> . 2024
	Jeremy Taylor, Kyuseong Choi , Peisong Han, “Data integration: exploiting ratios of parameter estimates from a reduced external model”, <i>Biometrika</i> . 2023
CONTRIBUTED PRESENTATIONS	Uniform inference for stationary distribution of diffusion process: wild regenerative block bootstrap for Harris recurrent Markov chains
	• 2025 JSM, Nashville Aug 2025
	• 2025 INFORMS, Atlanta Oct 2025
	Test bench and unified python package for Nearest Neighbor-based matrix completion
	• 2025 ICML CODEML Workshop, Seattle July 2025
	Supervised kernel thinning
	• 2024 Neurips, Vancouver Dec 2024
	Learning counterfactual distributions via kernel methods
	• 2025 ICCOPT, LA July 2025
	• 2024 INFORMS, Seattle Oct 2024
	• 2025 SCI, ACIC, Detroit May 2025
	Gaussian approximation in Reproducing Kernel Hilbert spaces
	• 2024 Workshop on Translational Research on Data Heterogeneity, St. Louis Apr 2024
	Robust data integration in generalized linear models
TEACHING	• 2023 JSM, Toronto Aug 2023
	TA for statistics, ML, and optimization courses at Cornell Tech, Cornell University, and University of Michigan. 2019–2024